

# Position Generalization in Translated FashionMNIST

Architecture, patch scale, and patch embedding

Final English report

12 fitted models / 24 test evaluations

**Abstract.** Under one controlled recipe, transfer is asymmetric: model baselines trained on A change by at most 0.73 points at center, while those trained on B lose 57.74-75.58 points at random positions. CNN leads all four settings (35.40% on B -> A). For ViT, patch size 8 leads three settings; Conv2d and Linear projections differ by at most 0.61 points in this single-seed run.

## 1 Question

A classifier may learn class appearance and object location together. We isolate location by placing each 28 x 28 image either uniformly over all valid integer positions (A) or at the canvas center (B).

| Setting | Training | Testing | Purpose                              |
|---------|----------|---------|--------------------------------------|
| A -> A  | random   | random  | in-distribution reference            |
| B -> B  | center   | center  | fixed-position reference             |
| A -> B  | random   | center  | transfer within A's position support |
| B -> A  | center   | random  | one-to-many position transfer        |

## 2 Experimental process

Each of six configurations is fitted once on A and once on B. Validation selects the checkpoint; both test distributions are then measured on the same 10,000 held-out images, producing 12 fits and 24 final evaluations.

| Item      | Controlled choice                                                       |
|-----------|-------------------------------------------------------------------------|
| Data      | 28 x 28 FashionMNIST image on a 64 x 64 black canvas                    |
| Split     | 54,000 train, 6,000 validation; official 10,000-image test set held out |
| Training  | 15 epochs; batch 64; AdamW, lr 1e-3, weight decay 1e-4                  |
| Selection | best validation checkpoint                                              |
| Reporting | top-1 test accuracy; elapsed training time; seed 42                     |

**Transfer direction.** B -> A expands from one observed location to 1,369 valid locations. A -> B evaluates one location already inside the support of A.

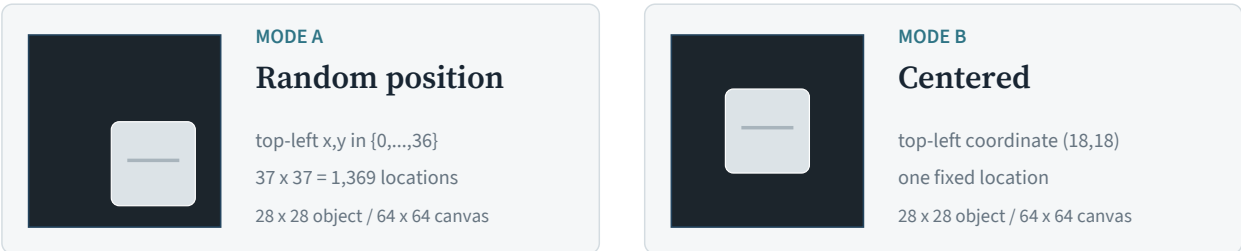


Figure 1. Mode A placements are deterministic per sample and seed; the same test placement seed is used for every configuration.

# Learning Curves and Checkpoints

Validation behavior for the three architecture baselines over 15 epochs

### 3 Training dynamics

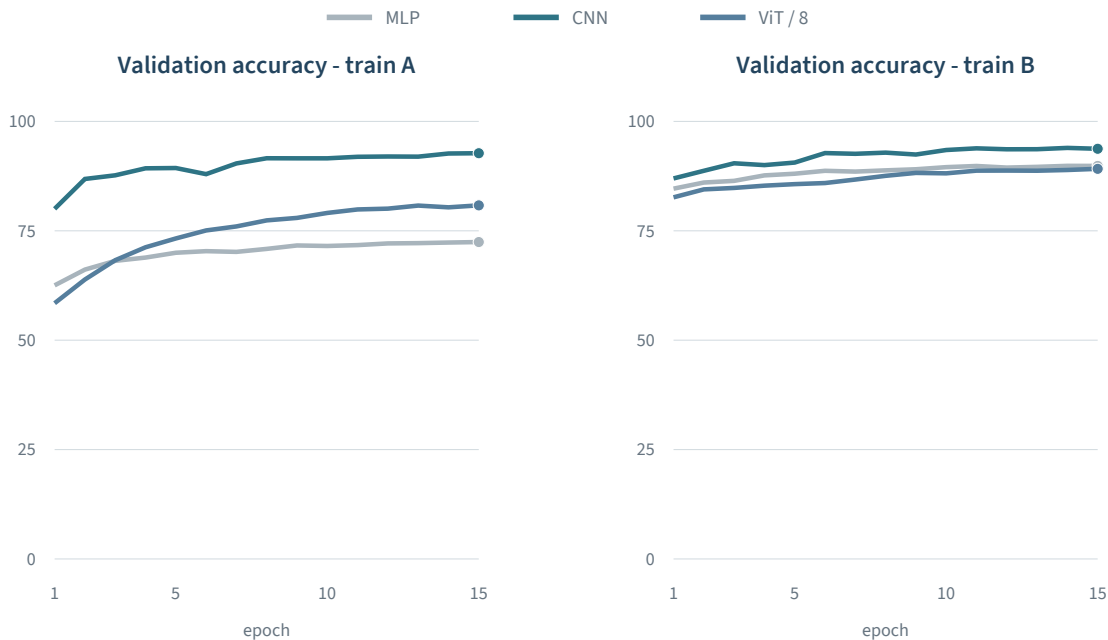


Figure 2. Validation accuracy for A-trained and B-trained architecture baselines.

| Configuration | Train A           | Train B           |
|---------------|-------------------|-------------------|
| MLP           | 72.43% (epoch 15) | 89.85% (epoch 14) |
| CNN           | 92.73% (epoch 15) | 93.95% (epoch 14) |
| ViT, patch 8  | 80.82% (epoch 15) | 89.17% (epoch 15) |

Table 1. Best validation accuracy and selected checkpoint epoch.

**Checkpoint evidence.** Every selected checkpoint occurs at epoch 14 or 15. CNN gives the strongest validation accuracy on both position distributions; all three models validate higher on the centered distribution.

| SELECTED EPOCHS           | BEST VALIDATION | LARGEST A/B GAP |
|---------------------------|-----------------|-----------------|
| 14-15                     | 93.95%          | 17.42 pp        |
| all six architecture fits | CNN / train B   | MLP validation  |

# Architecture and Position Shift

MLP (2 x 128), CNN (5 conv layers), and ViT (d=128, L=4, H=4, p=8)

## 4 Architecture results

| Configuration | Parameters | A -> A | B -> B | A -> B | B -> A |
|---------------|------------|--------|--------|--------|--------|
| MLP           | 542,474    | 71.89  | 89.57  | 71.97  | 13.99  |
| CNN           | 205,994    | 92.35  | 93.14  | 92.87  | 35.40  |
| ViT, patch 8  | 811,402    | 80.40  | 88.27  | 81.13  | 18.81  |

Table 2. Top-1 test accuracy (%); teal values mark column leaders.

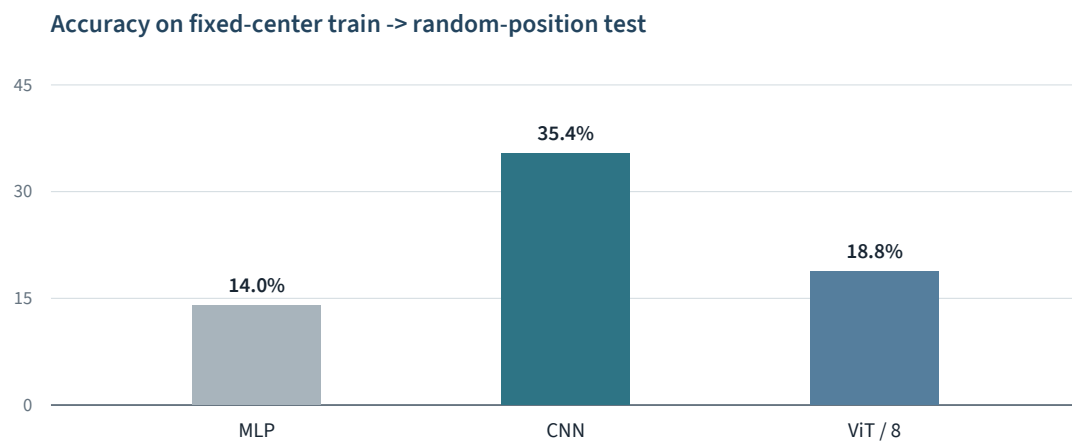


Figure 3. CNN reaches 35.40% on B -> A, 16.59 points above patch-8 ViT.

|                                                 |                                                                              |                                                                          |
|-------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>B -&gt; A LEADER</b><br><b>35.40%</b><br>CNN | <b>SMALLEST B -&gt; A DROP</b><br><b>57.74 pp</b><br>CNN, relative to B -> B | <b>A -&gt; B CHANGE</b><br><b>0.73 pp max</b><br>all three architectures |
|-------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------|

## 5 Position-shift analysis

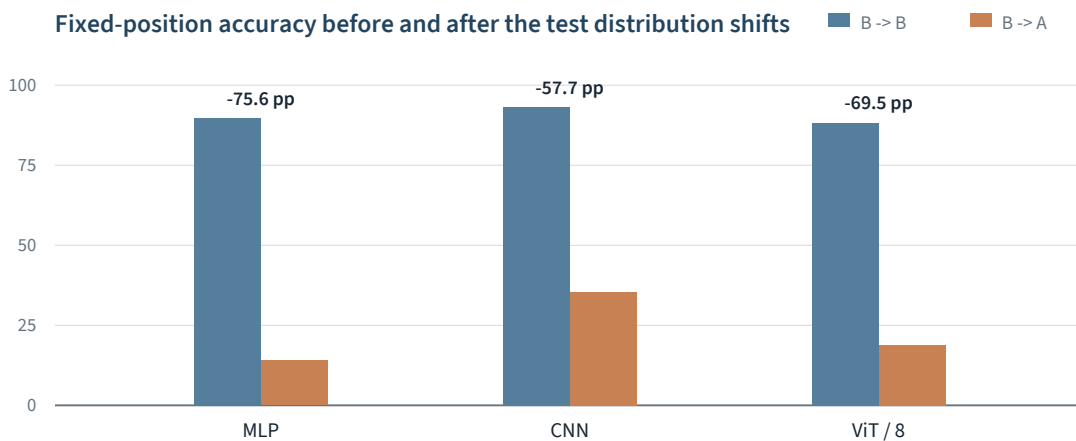


Figure 4. Centered-only training loses 57.74-75.58 points under random-position testing.

# Patch Scale and Embedding

ViT body fixed at  $d=128$ , depth 4, four heads, and FFN width 512

## 6 Patch-scale results

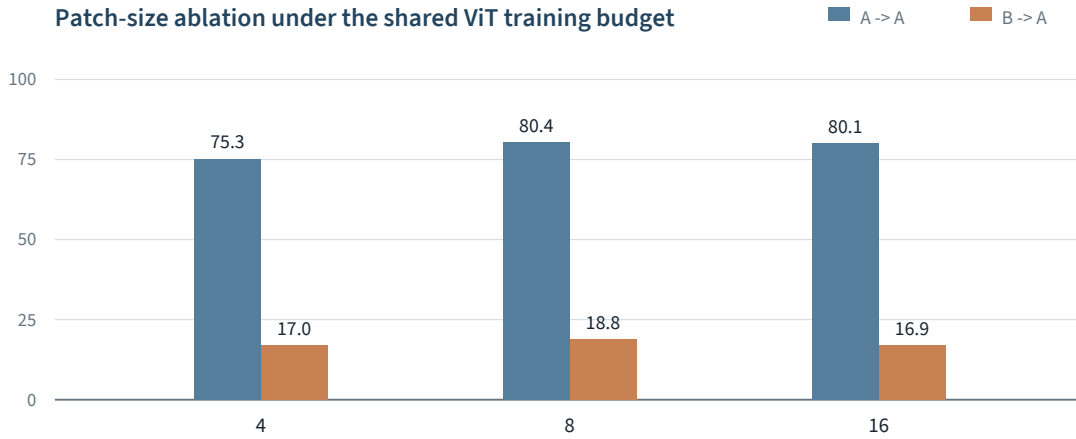


Figure 5. Patch size 8 leads three of four evaluated settings.

| Patch | Tokens | A -> A       | B -> B       | A -> B       | B -> A       | Time            |
|-------|--------|--------------|--------------|--------------|--------------|-----------------|
| 4     | 256    | 75.28        | 87.22        | 74.97        | 17.02        | 8.93 min        |
| 8     | 64     | <b>80.40</b> | 88.27        | <b>81.13</b> | <b>18.81</b> | 7.36 min        |
| 16    | 16     | 80.07        | <b>88.87</b> | 80.97        | 16.91        | <b>6.46 min</b> |

Table 3. Accuracy (%) and summed training time; teal marks leaders and fastest time.

**Result.** Patch size 8 leads on A -> A, A -> B, and B -> A. Patch size 4 raises the token count to 256 and is slower without an accuracy gain.

## 7 Patch-embedding results

For non-overlapping patches, Conv2d with  $\text{kernel} = \text{stride} = p$  and a shared Linear layer on each flattened  $p \times p$  patch implement the same affine map when their weights are aligned. Their trained results remain close.

| Test   | Conv2d | Flatten + Linear | Absolute gap |
|--------|--------|------------------|--------------|
| A -> A | 80.07  | 79.67            | 0.40         |
| B -> B | 88.87  | 88.80            | 0.07         |
| A -> B | 80.97  | 80.48            | 0.49         |
| B -> A | 16.91  | 16.30            | 0.61         |

Table 4. Test accuracy (%) for patch size 16. Maximum absolute gap: 0.61 points.

# Integrated Findings

Evidence-based interpretation with explicit limits

## 8 Results and interpretation

| Comparison       | Observed evidence                                                          | Interpretation                                                                                                         |
|------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Architecture     | CNN leads all four settings; B -> A = 35.40%                               | Local shared filters are a plausible advantage, but architecture and parameter count are not independently controlled. |
| Position support | A -> B changes at most 0.73 pp; B -> A loses 57.74-75.58 pp                | Training on broad position support transfers to center; center-only training does not cover random positions.          |
| Patch scale      | Patch 8 leads three settings; patch 16 is fastest; patch 4 uses 256 tokens | Patch 8 gives the strongest accuracy-cost balance under the shared 15-epoch budget.                                    |
| Patch embedding  | Maximum Conv2d vs Linear gap = 0.61 pp                                     | Equivalent affine projections produce numerically close results; the remaining gap is not a significance claim.        |

**Main answer.** High centered-position accuracy does not imply translation robustness. Exposure to varied positions during training is the decisive difference between the two transfer directions in this experiment.

## 9 Conclusions

- 01** **Transfer is strongly asymmetric.** A-trained models stay within 0.73 points at center, while B-trained models lose 57.74-75.58 points at random positions. CNN leads all four settings.
- 02** **Patch size 8 leads three of four settings.** It gives the best B -> A result without the 256-token cost of patch size 4.
- 03** **Equivalent embeddings stay close in this run.** Conv2d and Flatten + Linear differ by at most 0.61 percentage points.

## 10 Limitations

### Scope

- One training and placement seed; no variance estimate.
- Model sizes and optimal hyperparameters are not matched.
- Only translation on a black canvas is tested.
- Runtime is hardware- and order-specific.

### Interpretation rule

Small differences are described only as numerically close in this run; no significance claim is made. Conclusions apply to the shared protocol rather than all possible training recipes.